

KTechlab: An Integrated Development Environment

Being an integrated development environment (IDE) for PIC microcontrollers and electronics, KTechlab essentially enables you to do design and simulation tasks. It is an open source software released under the terms of General Public License v2 (GNU GPL). Try out KTechlab included in this month's EFY Plus DVD

SNEHA AMBASTHA

KTechlab presents an integrated environment for electronic design and automation (EDA) tasks. It is considered to be an educational tool to design and simulate circuits related to PIC microcontrollers and other electronics. KTechlab features an extensive circuit designer that helps auto-routing along with simulation of logic elements and common electronic components.

KTechlab features include:

1. Allows auto-routing and supports real-time simulation and debugging
2. Enables use of C, Assembly and high-level programming languages
3. Provides a pictorial format for microcontroller programming

Many-in-one design tool

KTechlab has several components well integrated to help designers. Some of these are described below:

Circuit Simulator. It can simulate linear devices, non-linear devices and logics too. KTechlab can further be integrated with external simulators like gspim so as to allow in-circuit simulation of PICs.

Editor. It can act as a schematic editor to provide a rich and real-time feedback of the simulation.

Visual Editor. It has a flowchart editor that allows KTechlab to construct PIC programs visually. This makes it easy to program a microcontroller. The editor provides a microcontroller with

all its ports defined as input or output. A designer can easily program that microcontroller by dragging arrows from those ports or by drawing a flowcode

for those ports.

For example, let us program a PIC16F84 microcontroller to blink an LED. To do that, we will have to set any one port as output. Let us do that for PORTB,0 pin. The two ways in which it can be programmed are:

1. Click the arrow on the PORTB,0 pin and drag it to the left.

2. Click 'advanced' on the microcontroller and set PORTB,0 pin as output by resetting last bit of te trist register of port B. Then draw the flowcode and save it.

Compiler. KTechlab has a high-level pro-

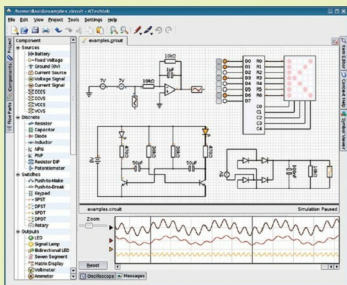


Fig. 1: KTechlab

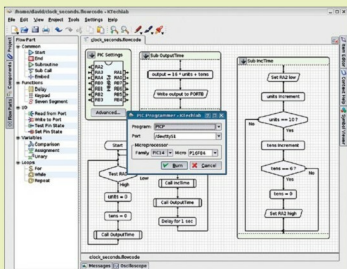


Fig. 2: Visual Editor